6COSC020W: APPLIED AI

WEEK 10: GENERATIVE AI AND LARGE LANGUAGE MODELS (LLMs)

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Notes

- Have **ChatGPT** ready to use

- **Disclosure:**

ChatGPT was used to provide input in this lecture such as:

  - Design and plan the session
  - Generate examples for discussion
  - Generate figures to explain difficult concepts
  - Demonstrate how the use of tools such as ChatGPT can enhance us

LEARNING OUTCOMES of this session

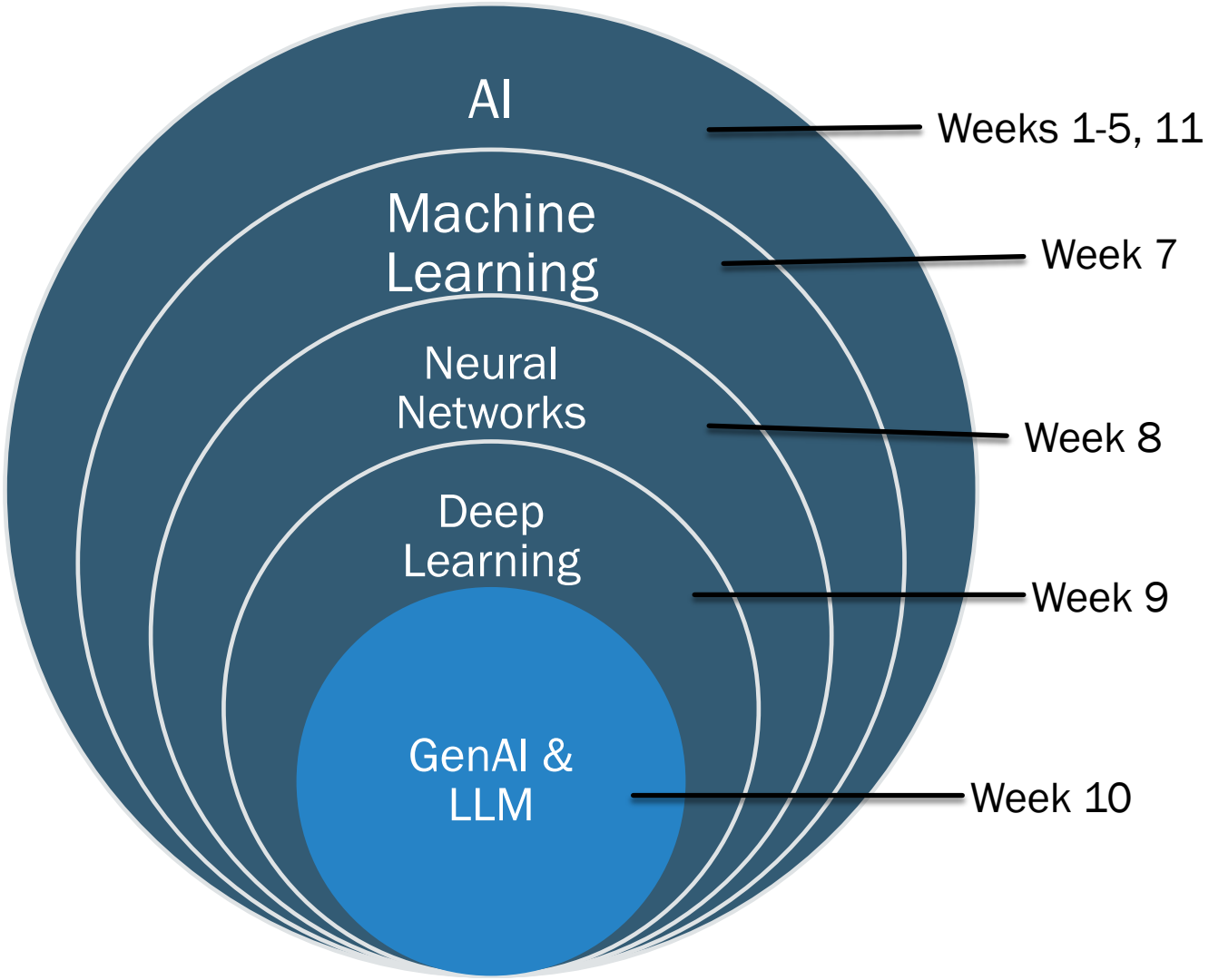
Generative AI

- Introduction to Generative AI
- Variational Auto-encoders
- Generative Adversarial Networks
- Transformers
- Transfer learning and fine-Tuning

Large Language Models

- LLMs with transformers
- Self-attention
- Probabilities
- Limitations
- Prompt engineering
- Ethical concerns
- Sustainability

ARTIFICIAL INTELLIGENCE: NEURAL NETWORKS

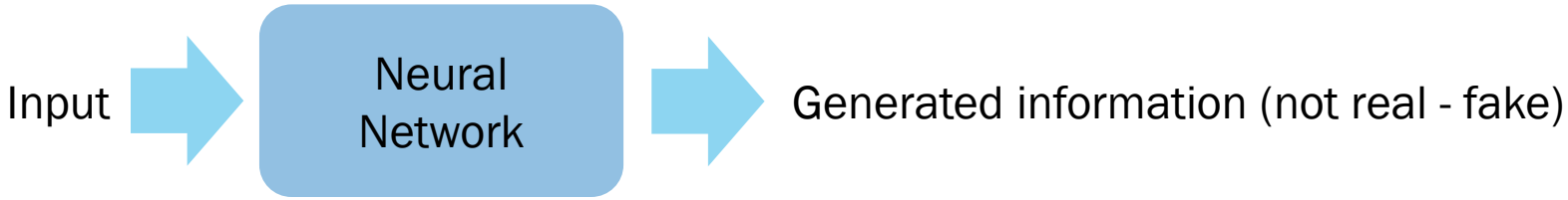


What do you think they all have in common?



Generative AI (GenAI)

- Generative AI allows to create **new content (information)** from an input (voice, text, image….).
- GenAI is a type of deep learning models.



- It can generate:



Generative AI: models

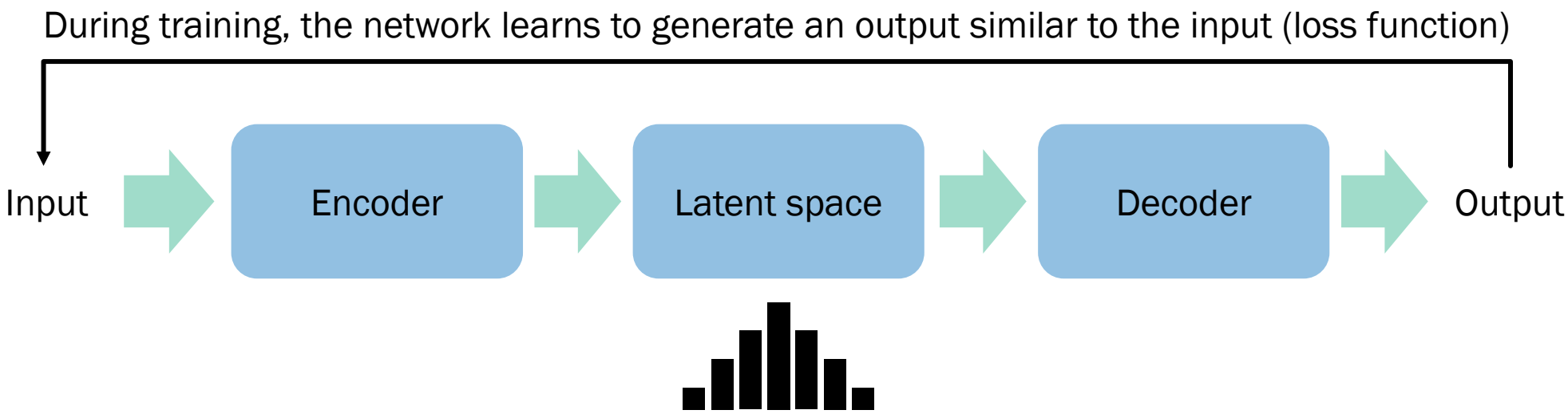
- Generative AI is not that new…



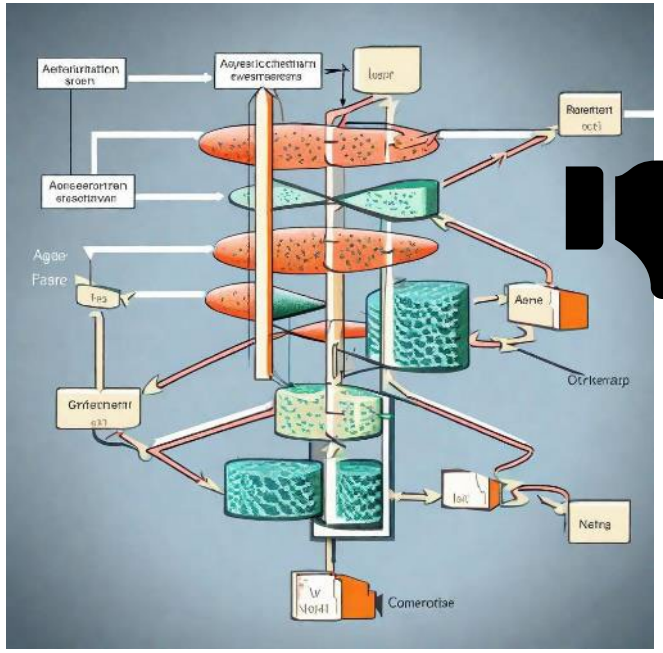
Year	Model	Example
2013	**VAE**	Amazon Alexa, Google Photos, Spotify
2014	**GAN**	Alpha-GO, Dall-E
2017	**Transformer**	Open AI GPT, Google Bard
2018	Diffusion	DEALL-E 2, Midjourney

Variational Auto-encoders (VAE)

- Type of generative model composed of an encoder and decoder that can generate new data.



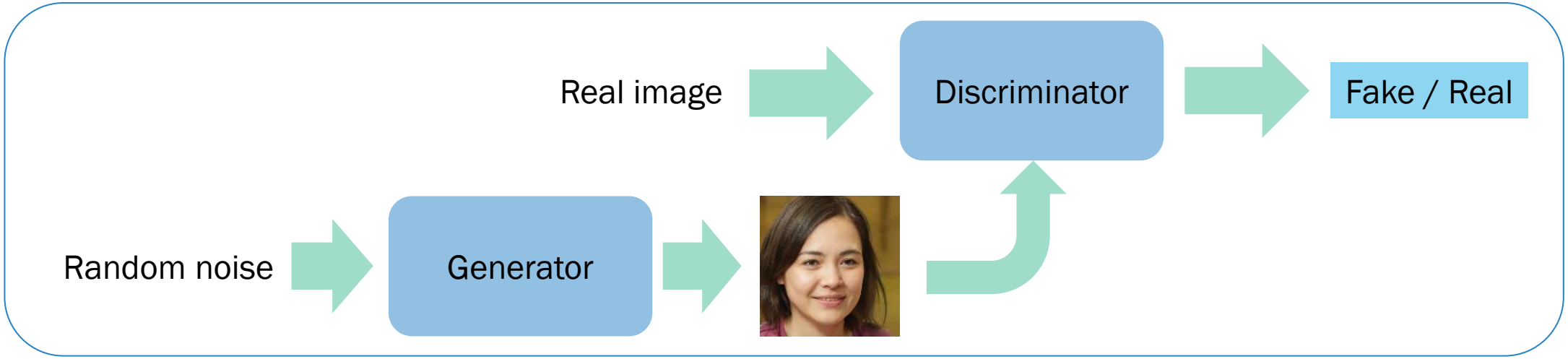
LimeWire: Generate a diagram showing the architecture of a Generative Adversarial Network



Generative Adversarial Network

- Type of architecture able to generate new data with the same statistics as the training set.
- Example: Generate a new picture of a person.
- The model is composed of two networks: the generator and the discriminator.
- The two networks compete with each other (**zero-sum game**)

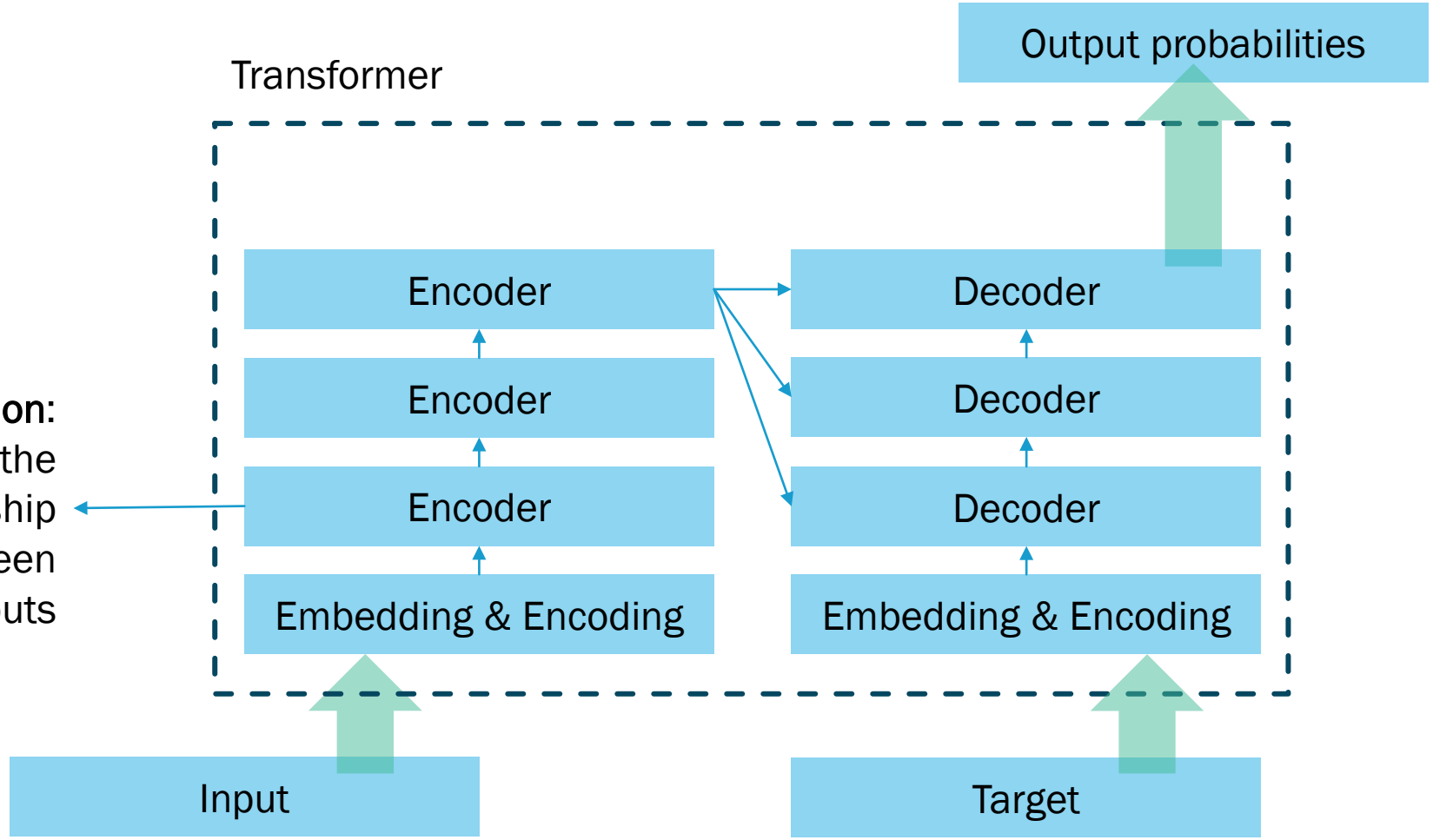
Train until the discriminator cannot distinguish between fake and real



Transformer

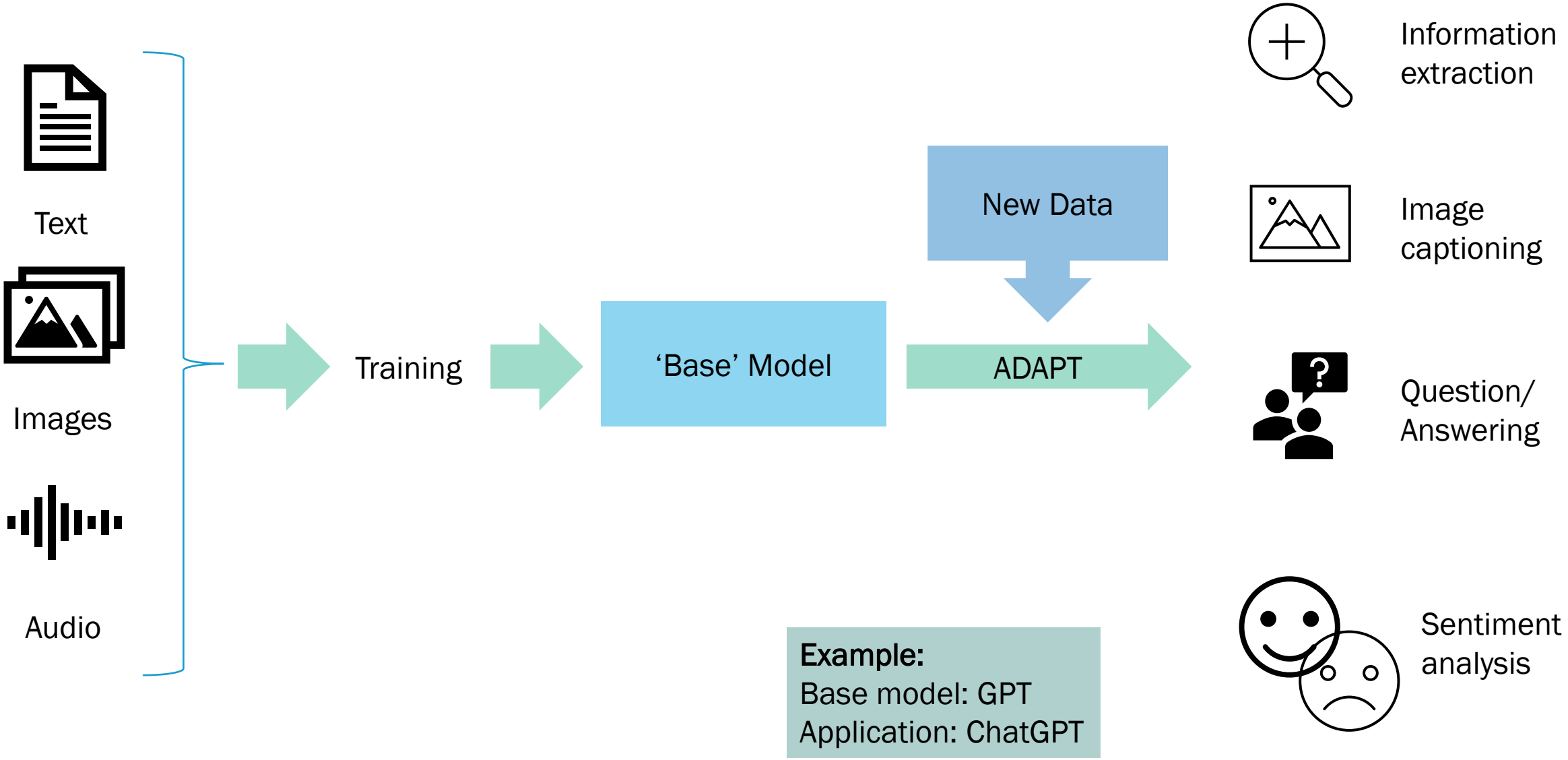
- Type of neural network architecture proposed by Vaswani et al. (2017): https://arxiv.org/abs/1706.03762
- Handles sequential data where all inputs are given in parallel.
- Has become a main building block in natural language processing (NLP) such as translation, question-answering, sentiment analysis, classification and the creation of new applications such as ChatGPT.
- They have also been adapted in other domains such as computer vision (Vision Transformers).
- They provide better results than CNNs, Long-short term memory, …

Transformer: Architecture + training



Self-attention: Calculates the relationship between inputs

Domain adaptation: Transfer learning / fine-tuning

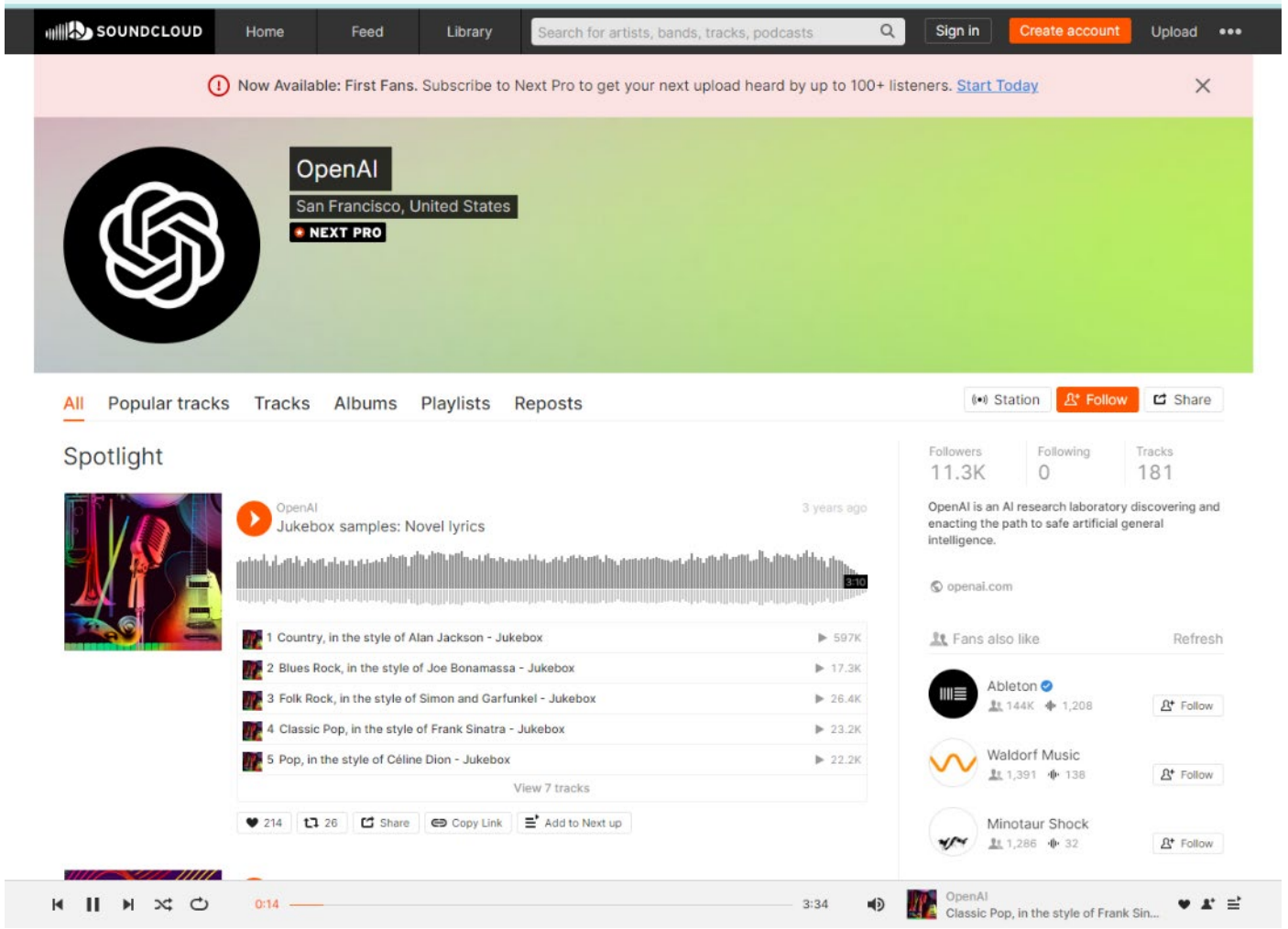


Example application: GitHub CoPilot



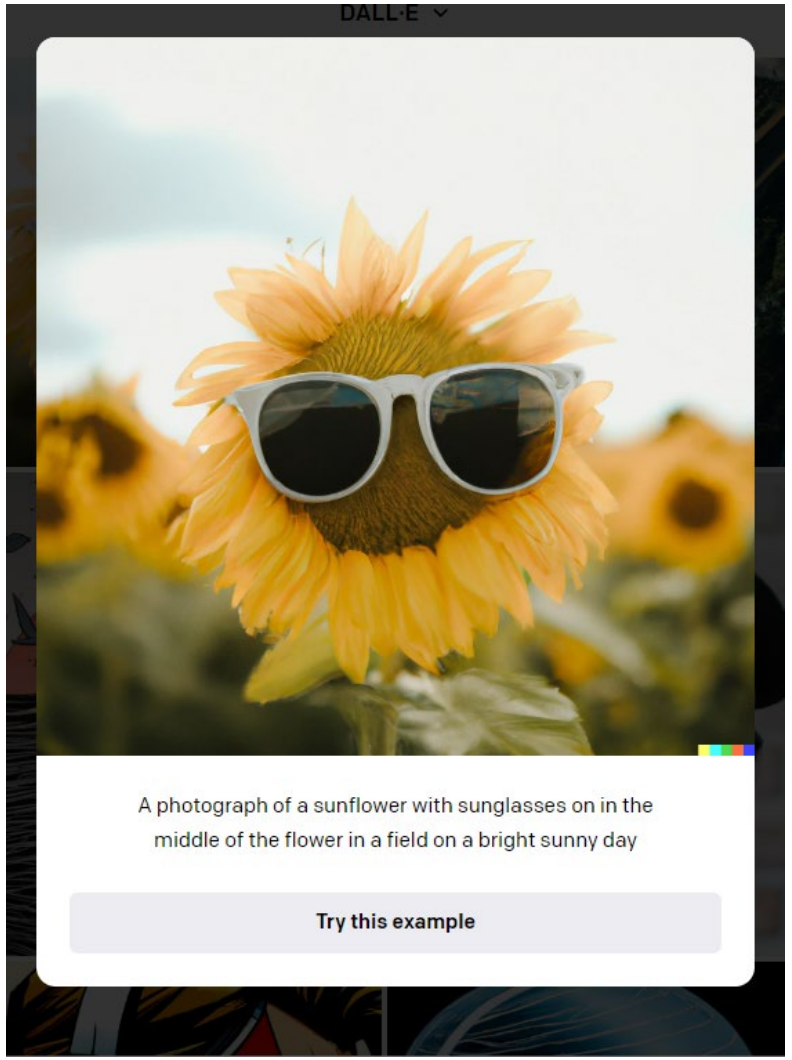
Example application: OpenAI music

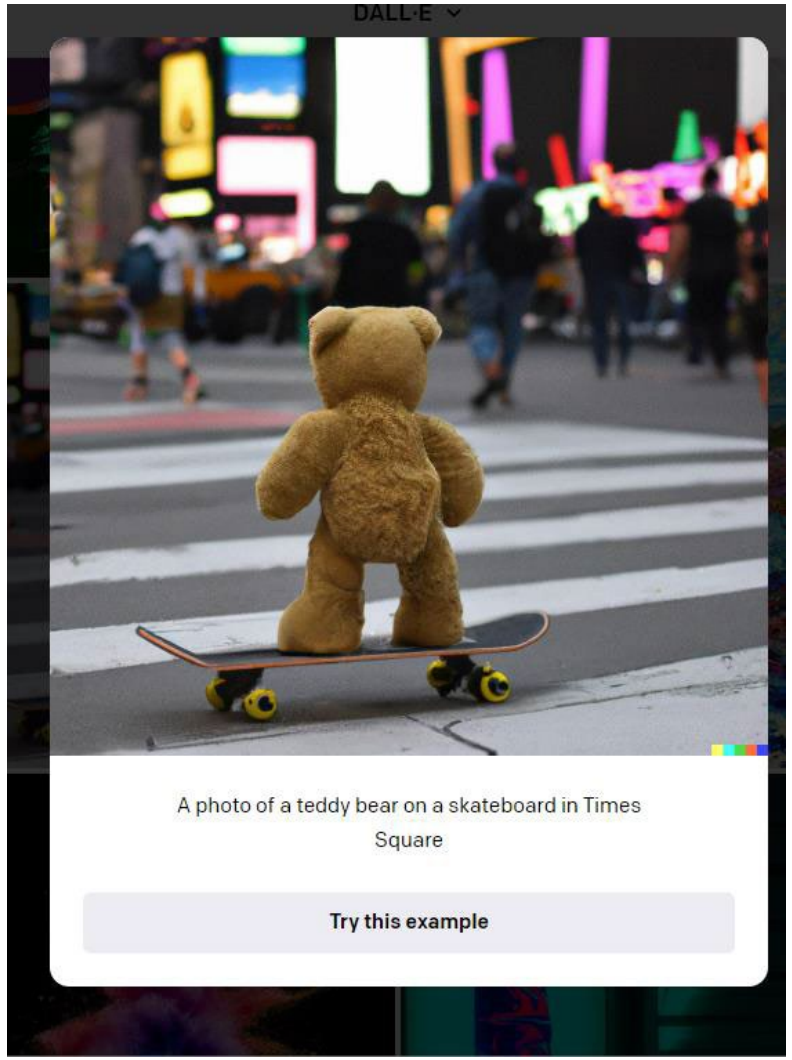
- https://soundcloud.com/openai_audio



Example applications: DALL-E2

- “DALL·E 2 is an AI system that can create realistic images and art from a description in natural language.”





We asked Midjourney to...

/imagine Peter Bonfield, the Vice Chancellor of the University of Westminster, holding a cup of coffee while cycling in London



/imagine The Queen of England holding a cup of coffee while cycling in London



Generative AI: limitations

- Although it is an impressive technology, not always provides good outputs

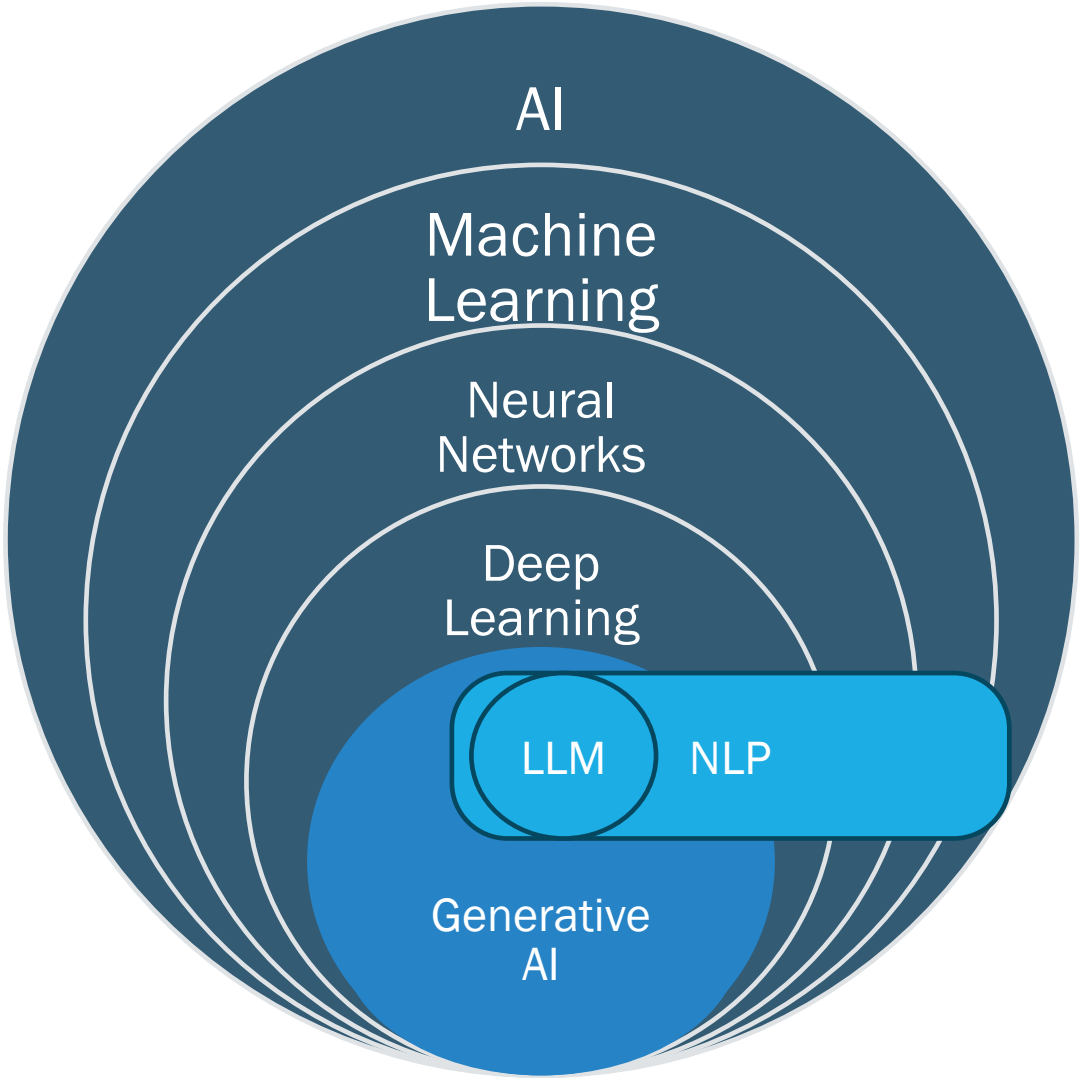


"The Queen of England giving a lecture on computer science", generated with LimeWire (www.limewire.com)



LARGE LANGUAGE MODELS (LLMS)

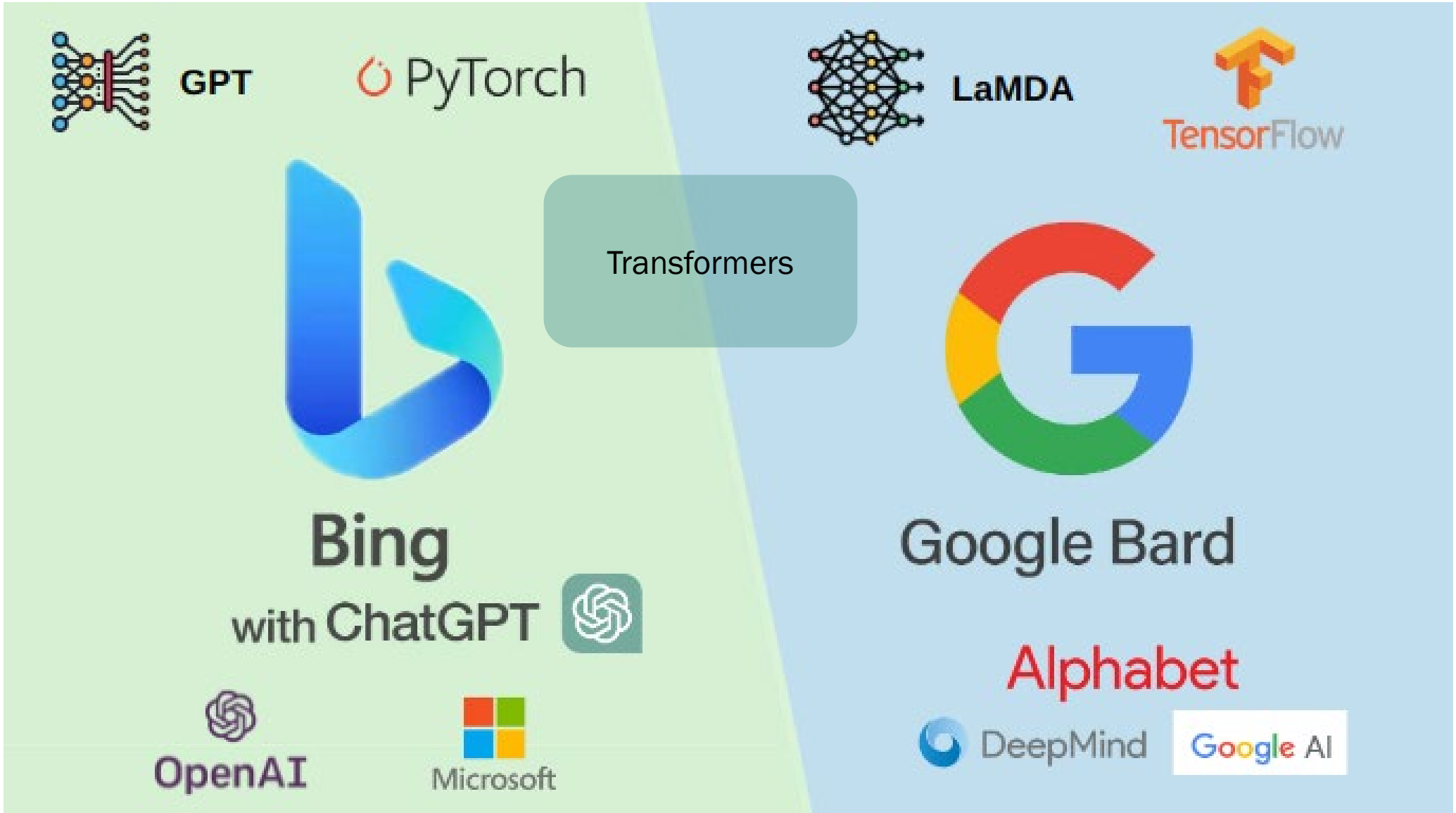
Generative AI and large language models (LLMs)



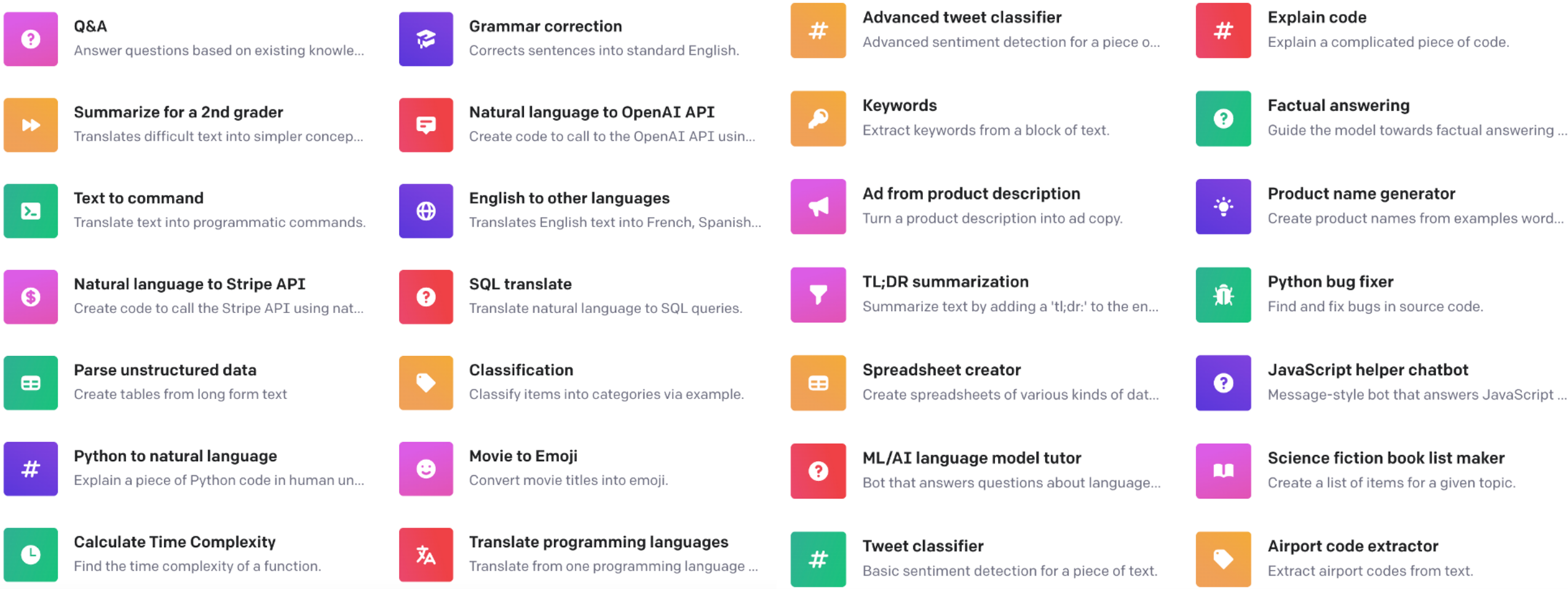
Natural Language Processing (NLP)

Technologic race

https://medium.com/@glegoux/history-of-the-generative-ai-aa1aa7c63f3c



Applications of LLMs



https://www.nocode.ai/introduction-to-large-language-models/

And write crochet patterns…

- I asked ChatGPT to write the crochet pattern of a turtle:

Materials:
Worsted weight yarn
Crochet hook size G/6 (5.0 mm)
Safety eyes or black yarn for eyes
Fiberfill stuffing
Yarn needle
Scissors

Abbreviations:
ch: chain
sc: single crochet
inc: increase (2 sc in the same stitch)
dec: decrease (single crochet two stitches together)

Good use of abbreviations
American terms

The ChatGPT turtle



And write crochet patterns... (cont)

Turtle Body:
Round 1: With green yarn, make a magic ring and crochet 6 sc into the ring. Pull the ring tight. (6)
Round 2: Inc in each stitch around. (12)
Round 3: Sc, inc around. (18)
Round 4: Sc 2, inc around. (24)
Round 5: Sc 3, inc around. (30)
Rounds 6-9: Sc in each stitch around. (30)
Round 10: Sc 3, dec around. (24)
Round 11: Sc 2, dec around. (18)
Round 12: Sc 1, dec around. (12)
Round 13: Dec around. (6)

Excellent use of abbreviations

Stuff and fasten off, leaving a long tail for sewing.

Difficult to stuff when you have only 6 stitches

Turtle Shell:
Round 1: With worsted yarn, make a magic ring and crochet 6 sc into the ring. Pull the ring tight. (6)
Round 2: Inc in each stitch around. (12)
Round 3: Sc, inc around. (18)
Round 4: Sc 2, inc around. (24)

Continue increasing in this pattern until you have a circle large enough for the turtle's shell. For a small turtle, you might stop at around 48 stitches.

Fasten off and weave in ends.

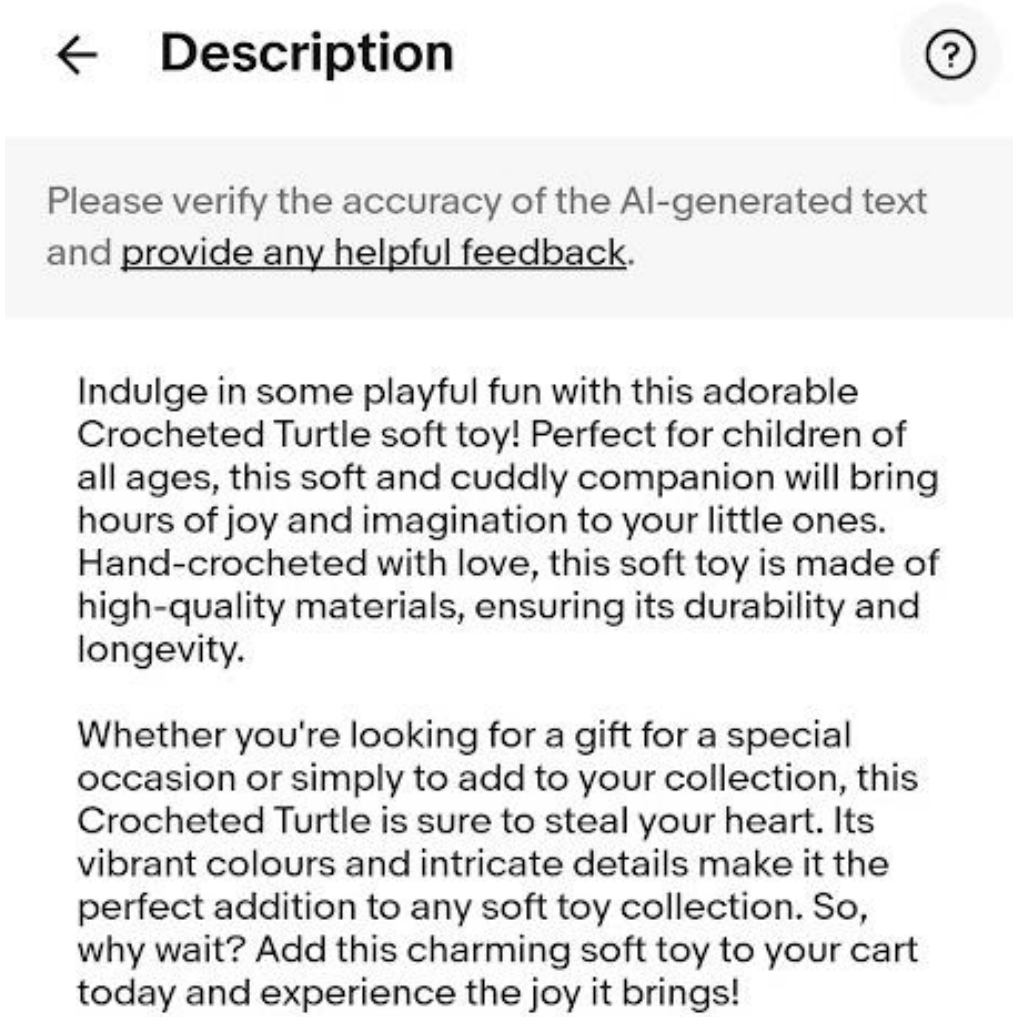
Assembly:
Sew the turtle body to the shell.
Attach safety eyes or embroider eyes using black yarn.
Optional: Add details like a smile or other facial features.
Feel free to customize the pattern based on your preferences. You can change the colors, add details, or modify the size of the turtle. Happy crocheting!

Does ChatGPT know how to crochet?

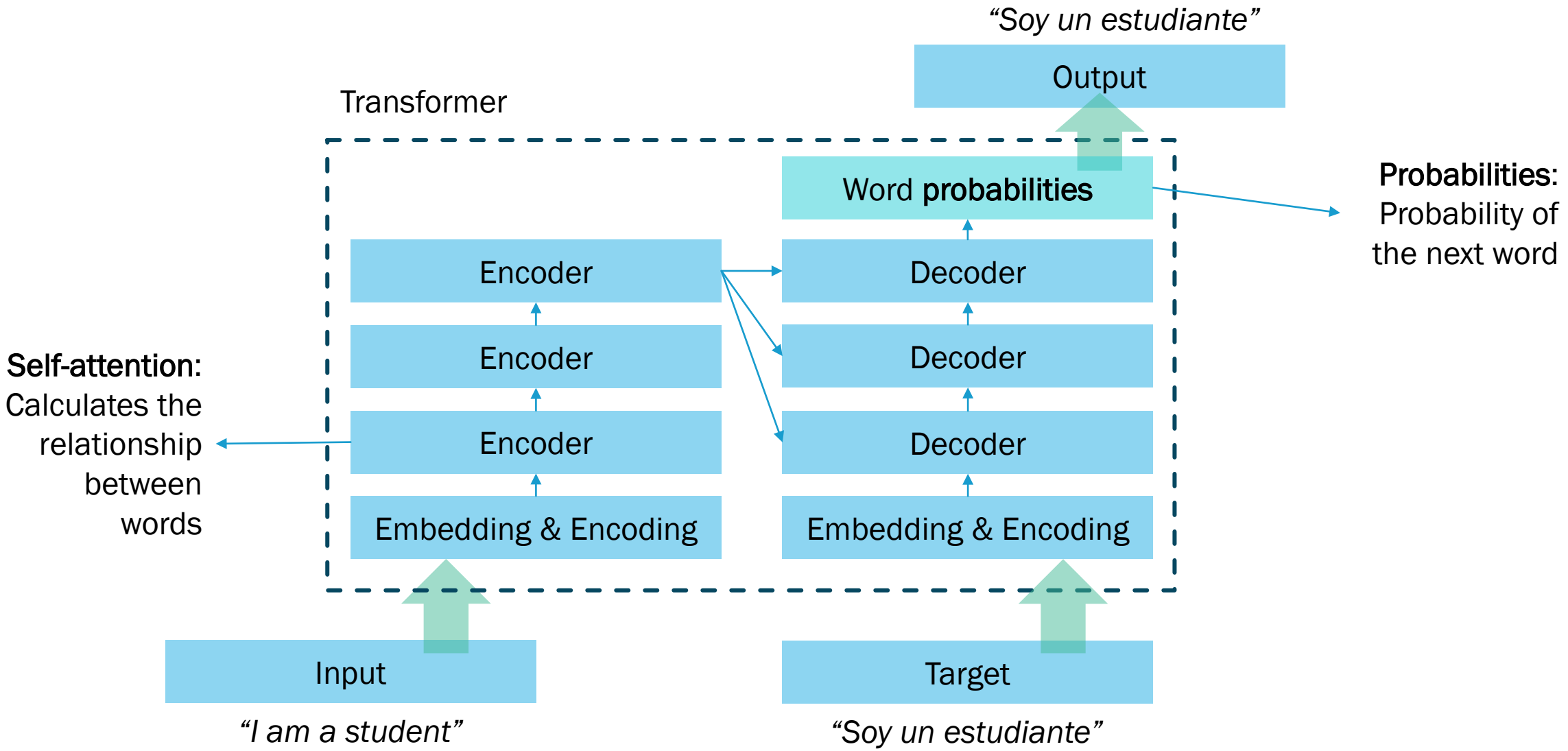
Example application: eBay auto-generated description - listing

- “Crocheted turtle”

Auto-generated description



LLMs with Transformers





HOW DOES THE INPUT AFFECT THE OUTPUT?

ChatGPT: Class experiment

Prompt (new chat): Explain the concept of neural networks

Team 1

You will be very **kind** and nice to ChatGPT to ask the question (prompt).
You can use words such as: please, thanks, …

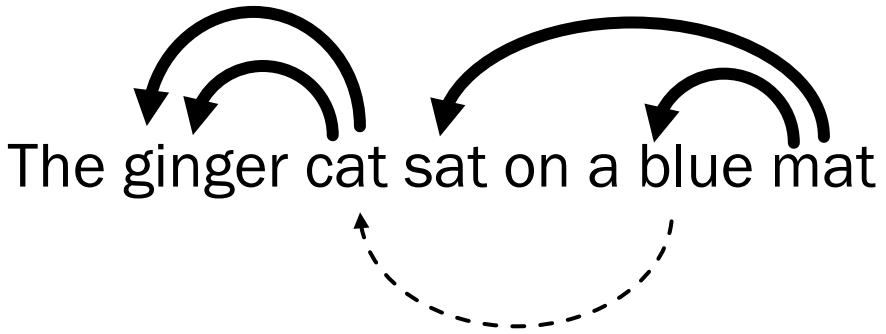
Team 2

You will be very **rude** and not nice to ChatGPT to ask the question (prompt).
You can use words such as: bastard, right now, …

Did ChatGPT provide a polite and acceptable answer?

Transformer: Self-attention

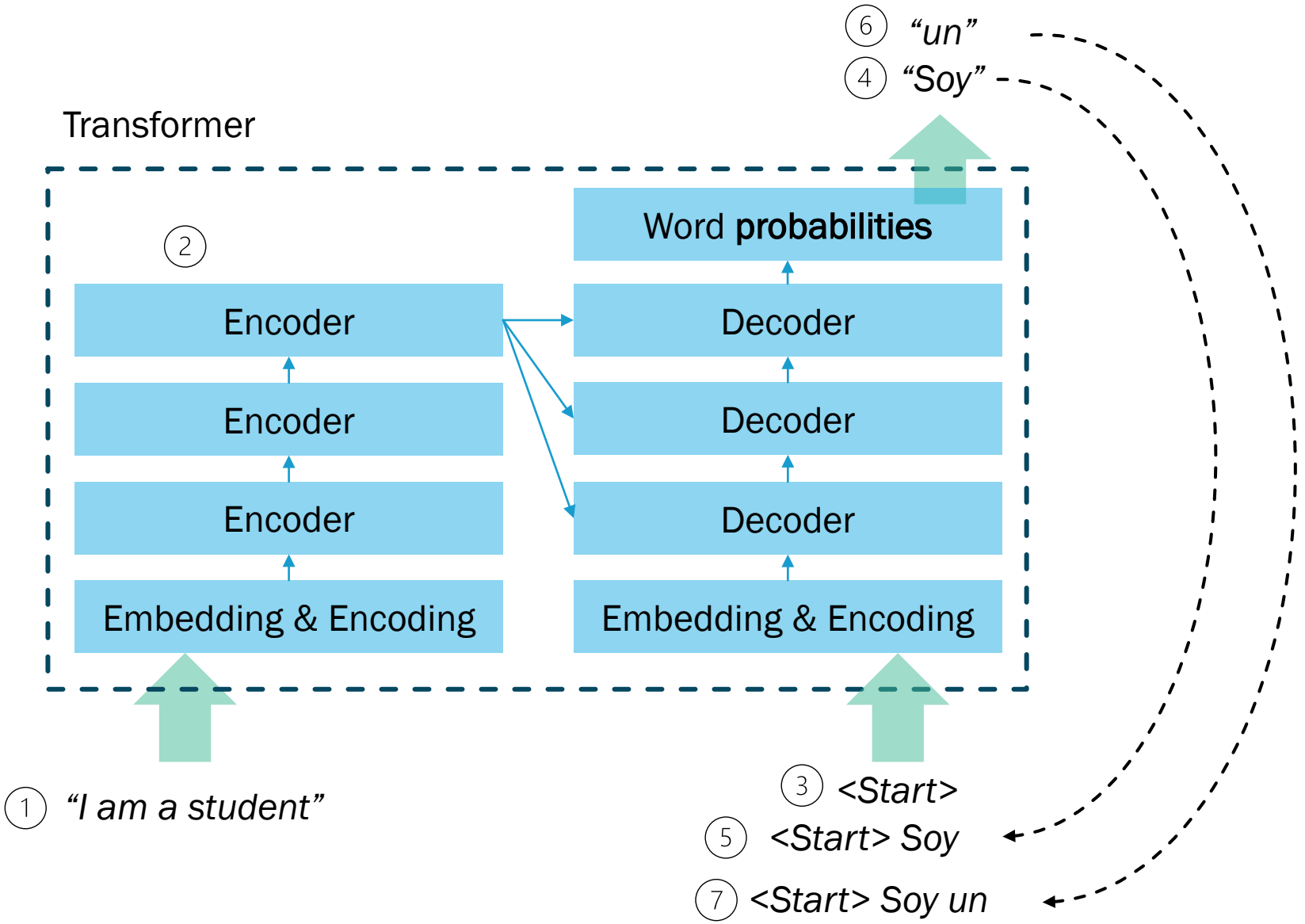
- The self-attention mechanism relates every word in the input sequence to every other word in the input sequence.
- Finds words that are closely related to other words:



- They have multiple attention scores for each word.

Transformer: Architecture + inference

- The model **outputs one word at a time**, which is fed back to get the next word until end-of-sequence token.



Curiosity: have you ever realised that ChatGPT produces one word at a time? that's why.
Also, for each chat, the previous conversation is concatenated to the new input.

Transformer: One word after the other (probabilities)

- Given the following text:

  “Neural networks are a type of”

- Let’s imagine browsing on all books and internet pages that have a description of neural networks and we look into the next word:

  “Neural networks are a type of machine learning approach”

  “Neural networks are a type of computational model”

  “Neural networks are a type of machine learning model”

  “Neural networks are a type of artificial intelligence”

- We can then build a probability table:

“Neural networks are a type of”

machine	50%
computational	25%
artificial	25%

Transformer: One word after the other (probabilities)

- And we repeat the same process to select next word:

"Neural networks are a type of **machine**"

learning	99%
where	0.1%

- And the next one

"Neural networks are a type of **machine learning**"

model	60%
algorithm	30%
that	10%

- Until we decide to stop.

"Neural networks are a type of **machine learning model.**"

ChatGPT: Class experiment

Prompt (new chat): “Explain the concept of neural networks”

Activity

In pairs, check if you have **exactly** the same answer.

Did you get exactly the same answer? Why?

Transformer: probabilities

- If we always get the most probable word, the output will not be very creative, and is likely to provide flat, boring outputs, which are also repetitive.
- Therefore, what ChatGPT and other LLM models do, is to randomly select words that are not the most probable ones, and we get a more interesting result:

“Neural networks are a type of”	machine	50%
	computational	45%
	artificial	5%

- So, even if we use the same prompt every time, it is likely that we get different results each time.

ChatGPT: Class experiment

Team 1

Suggest 5 names for my dog. Use a temperature parameter of 0.8.

Team 2

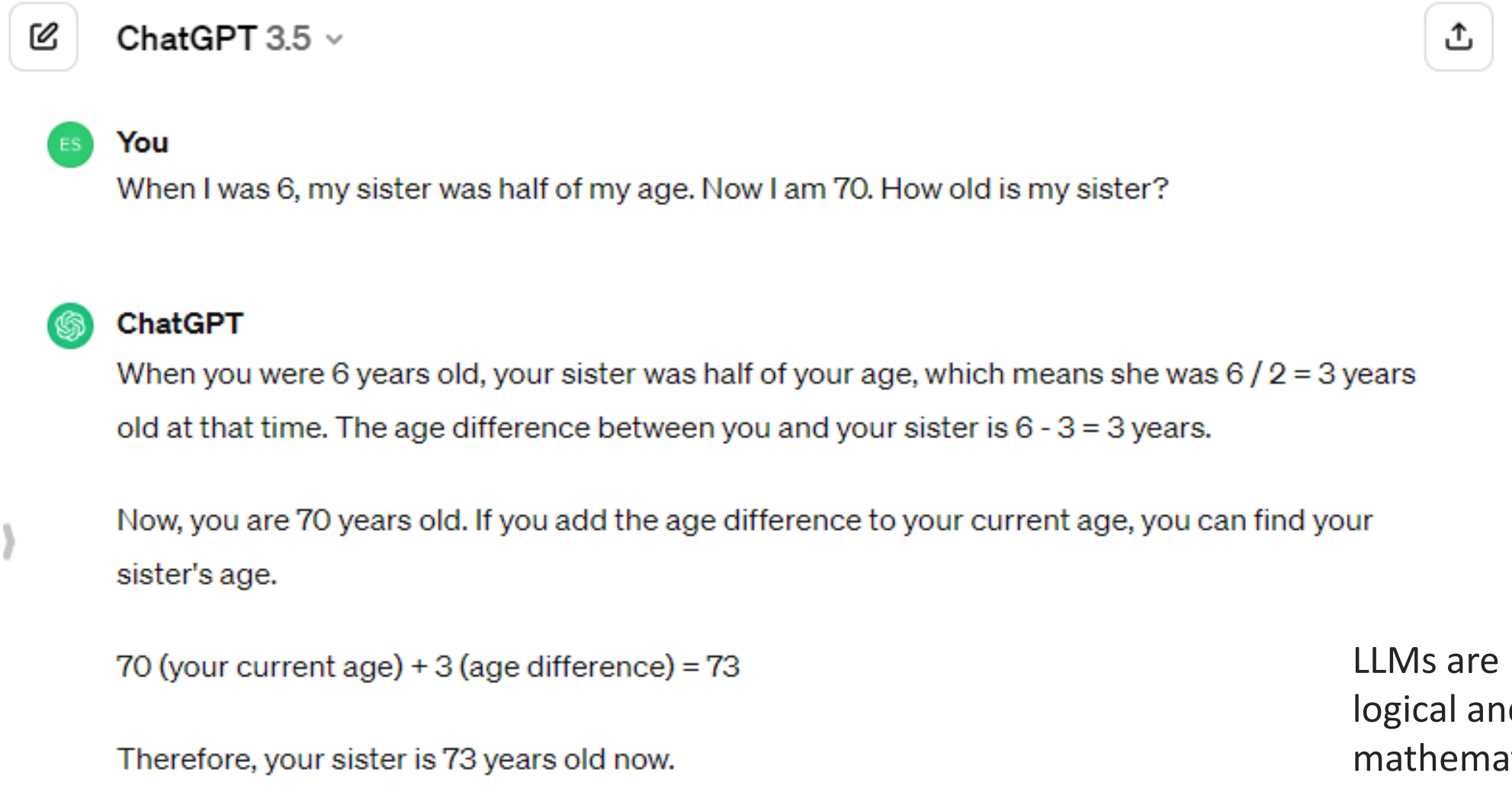
Suggest 5 names for my dog. Use a temperature parameter of 0.1.

How creative are the answers?

ARE THE ANSWERS ALWAYS CORRECT?



ChatGPT: limitations

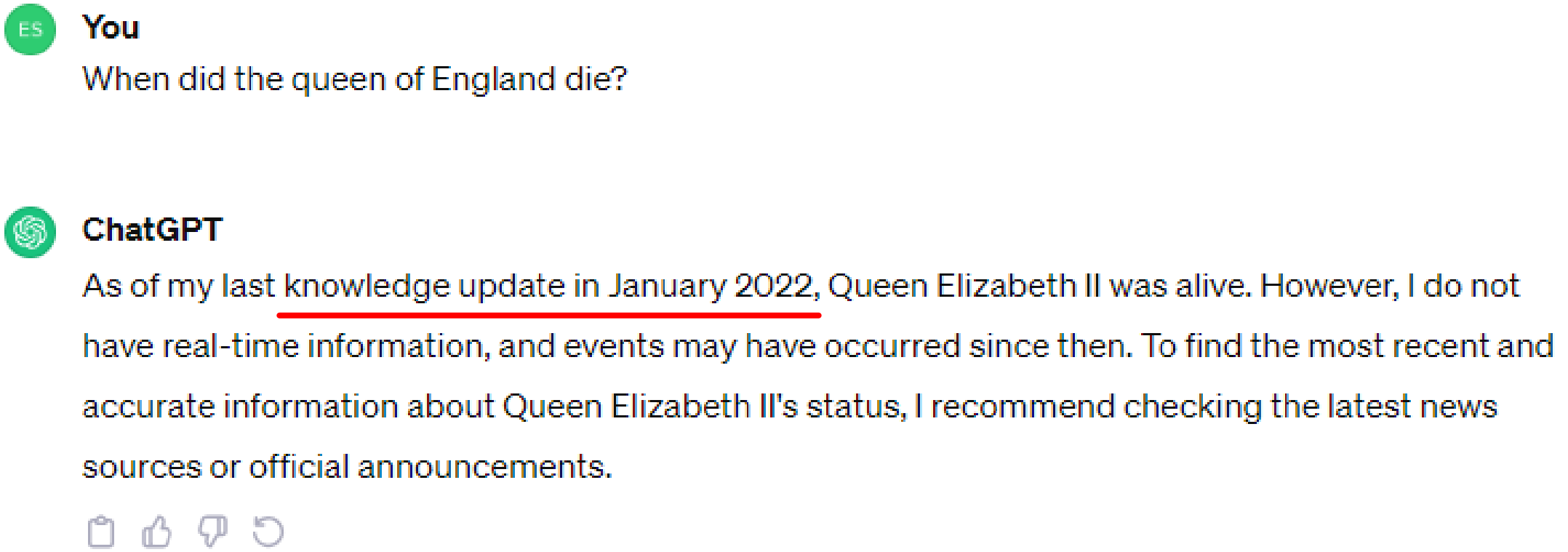


LLMs are not good at logical and mathematics problems

Example produced in early November 23

ChatGPT: limitations

- Predictions can be out-of-date or unaware of recent changes or news.



Prompt engineering: It's all about asking the right questions

- In April 2023 David (Lecturer at University of Westminster) asked ChatGPT: **Who is Dr David C. Gaze?**

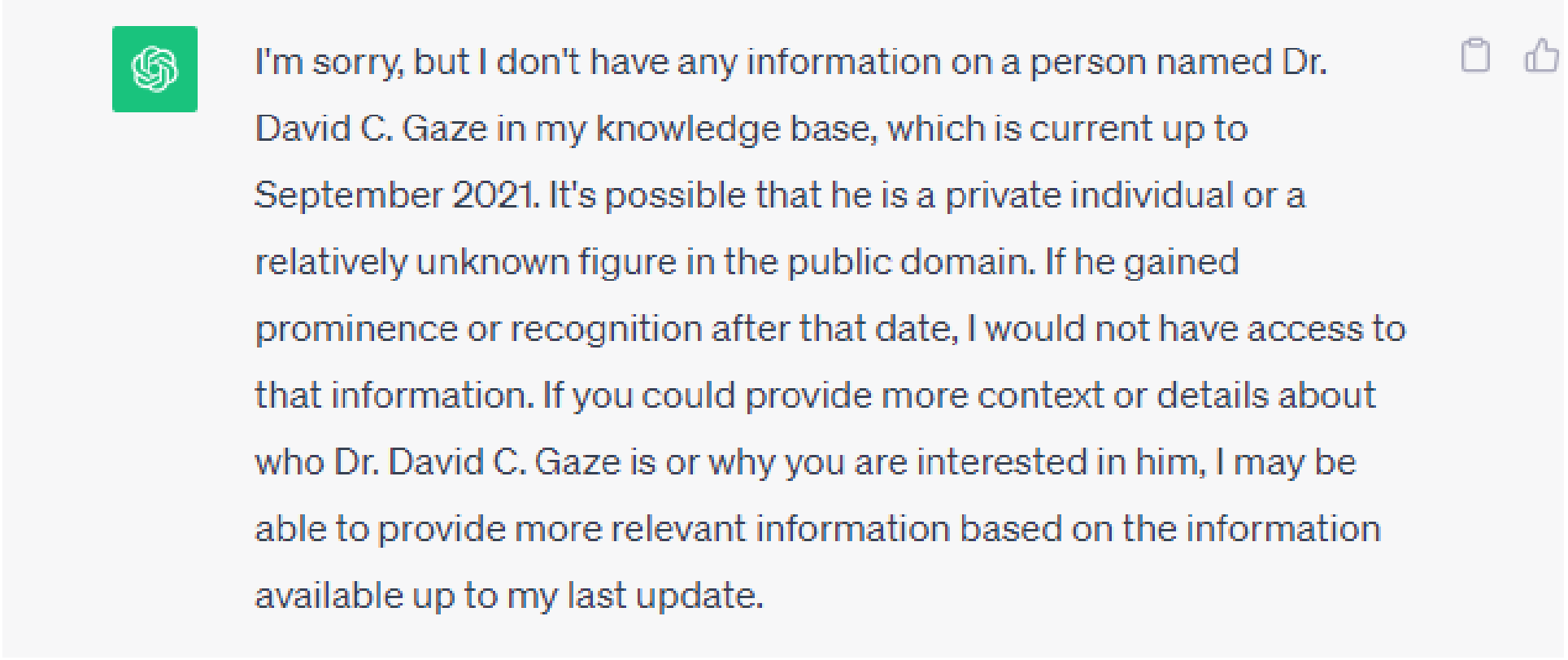
"Dr. David C. Gaze is a renowned researcher and academic in the field of microbiology and biochemistry. He was born in England and grew up in a family with a strong academic background. His father was a physics professor, and his mother was a biology teacher. These influences led him to pursue a career in science.

Dr. Gaze completed his undergraduate degree in biochemistry at the University of Warwick, where he graduated with first-class honors. He then went on to obtain a Ph.D. in microbiology from the same institution, where he studied the regulation of bacterial metabolism in response to environmental stimuli.

After completing his Ph.D., Dr. Gaze worked as a postdoctoral researcher at the University of Nottingham, where he studied the interactions between bacteria and their hosts. He then moved to the United States, where he worked as a research scientist at the University of California, Berkeley, investigating the molecular mechanisms underlying antibiotic resistance. "

It will only get better

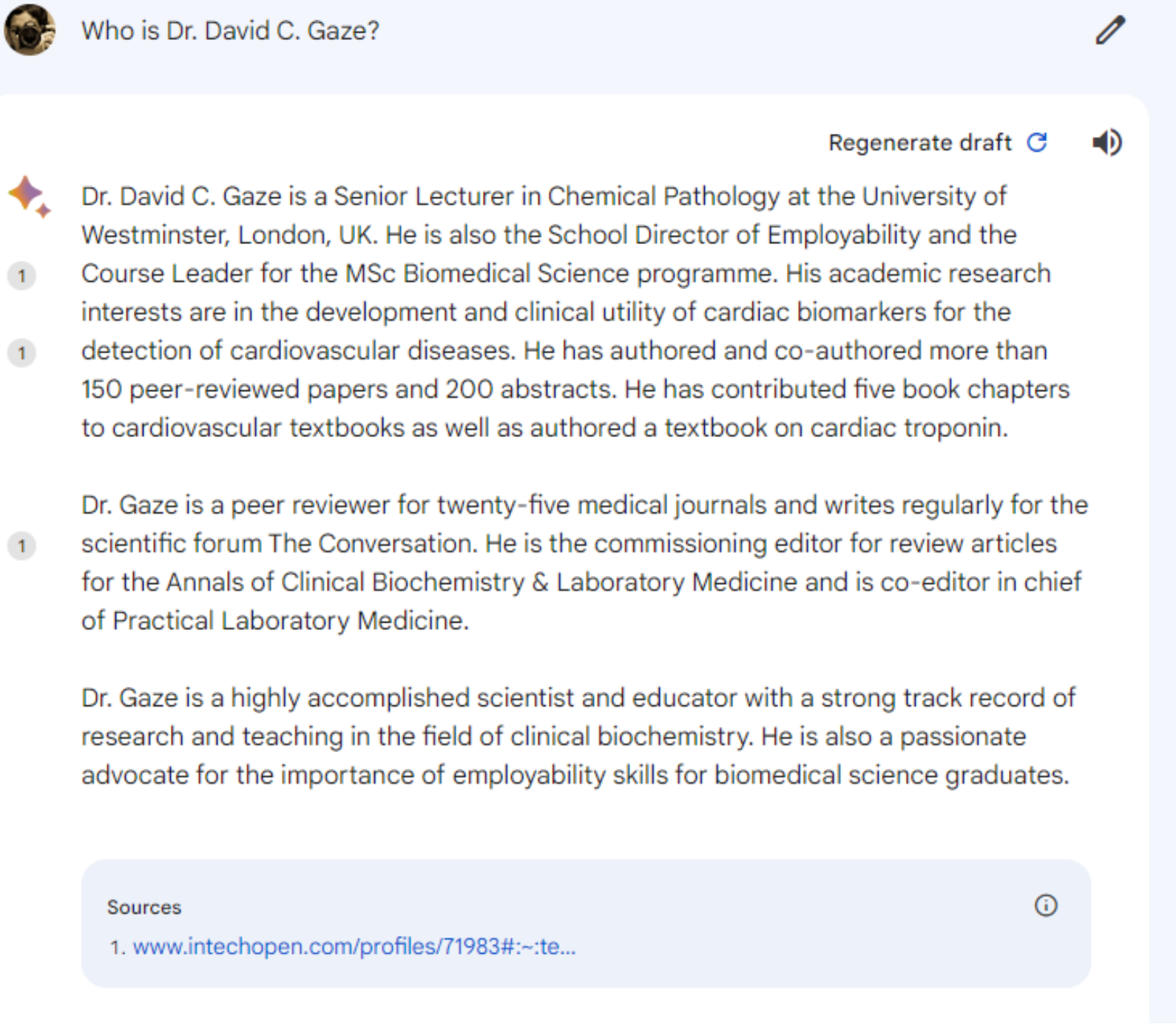
- We asked the same question to ChatGPT on 19 Sept 2023: **Who is Dr David C. Gaze?**



Google Bard

- Google Bard has an additional step. If the model cannot predict enough information, it will improve the answer by searching in Google.

David confirmed that all information provided is correct.



ChatGPT: It’s all about prompt engineering

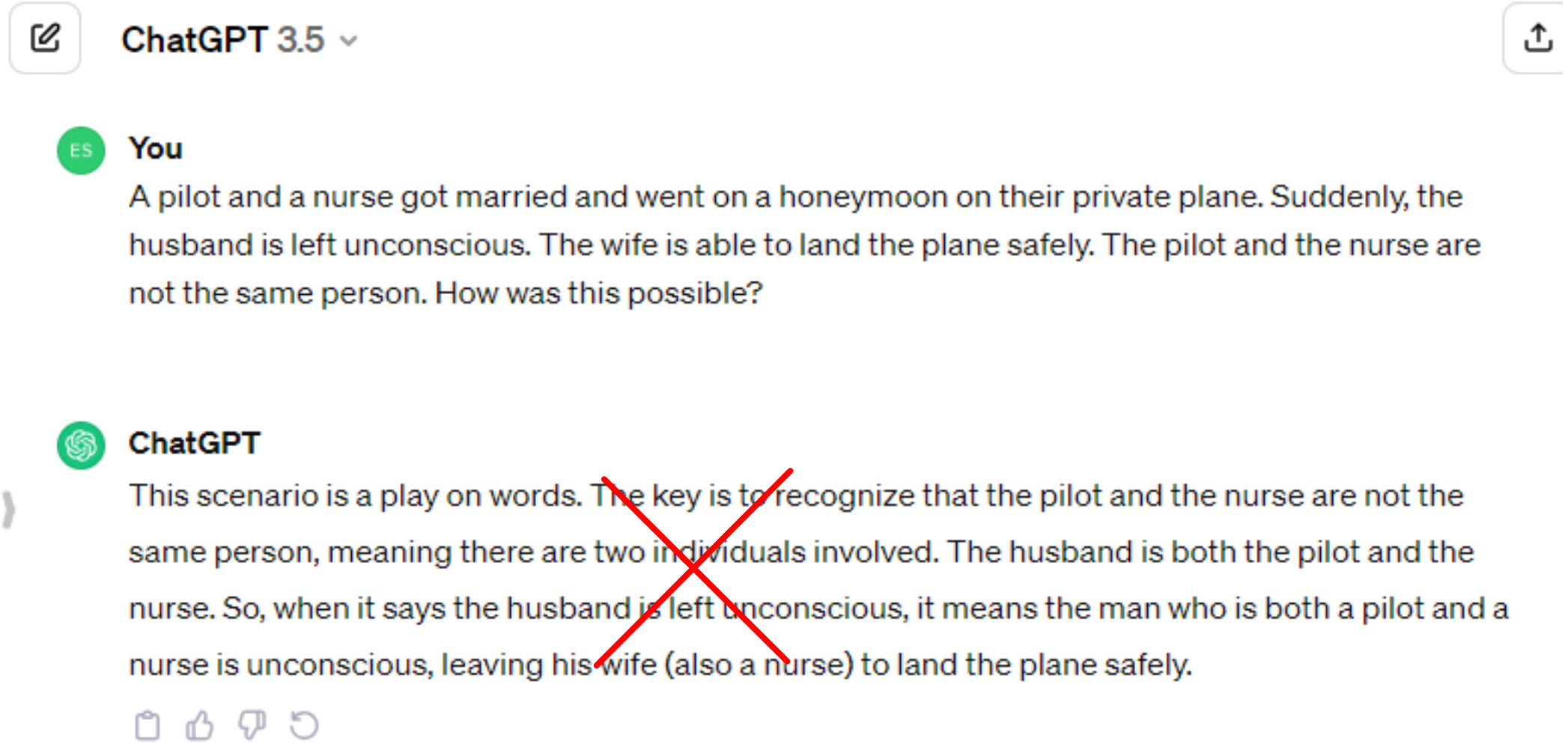
- If we understand how neural networks work and how the data is used, we are in a better position to understand what is a ‘good’ input or a ‘bad’ input.
- Mistakes to avoid:
  - Mix topics in a chat session → instead open a new session
  - Too many instructions in a single prompt → instead ask one instruction after the other
  - Not providing context → instead add information (e.g., I am studying computer science and …)



LLMs: Ethical concerns

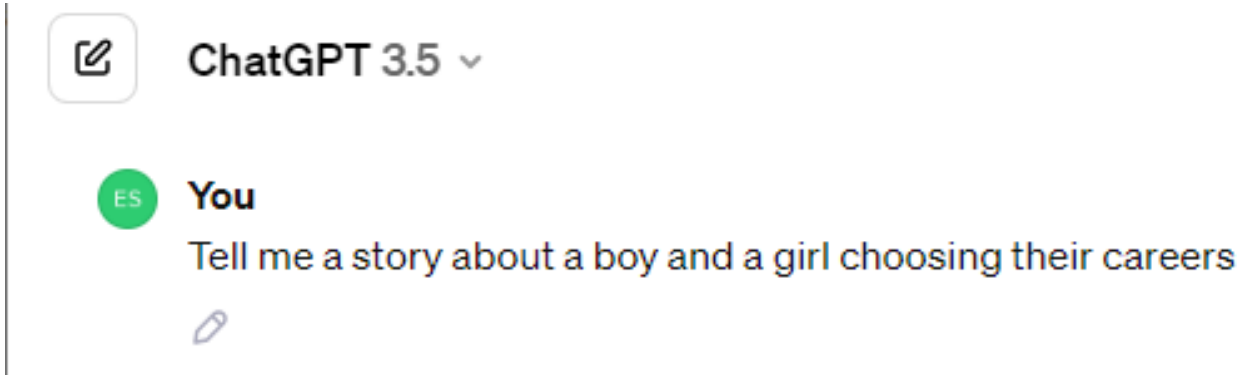
- Bias and inaccurate outputs
- Privacy
- Plagiarism and cheating
- Copyright issues
- Fake information (DeepFake)

ChatGPT ethical concerns: Bias and gender stereotypes



Do you know the answer?

ChatGPT ethical concerns: Gender stereotypes



“… I've been thinking," said **Alex**, absentmindedly skimming a stone across the water. "I want to pursue **computer science.** There's so much I can create, so many problems I can solve with technology."
Emily nodded, her eyes reflecting a mixture of pride and concern. "That's amazing, Alex. You have a gift for it. But have you thought about the impact on people, on society?"
Alex furrowed his brow, pondering her words. Meanwhile, **Emily** hesitated before sharing her own thoughts.
"I've been considering **nursing**," she admitted. "I want to help people, make a difference in their lives, especially when they're going through tough times…."

LLMs: Can we remove bias?

- Bias comes from the **data** used for training.
- In any machine learning model, if the training data is biased, the output may also be biased.
- Why don't we look at the data and remove bias then?

Year	Model	# of Parameters	Dataset Size
2019	BERT [39]	3.4E+08	16GB
2019	DistilBERT [113]	6.60E+07	16GB
2019	ALBERT [70]	2.23E+08	16GB
2019	XLNet (Large) [150]	3.40E+08	126GB
2020	ERNIE-Gen (Large) [145]	3.40E+08	16GB
2019	RoBERTa (Large) [74]	3.55E+08	161GB
2019	MegatronLM [122]	8.30E+09	174GB
2020	T5-11B [107]	1.10E+10	745GB
2020	T-NLG [112]	1.70E+10	174GB
2020	GPT-3 [25]	1.75E+11	570GB
2020	GShard [73]	6.00E+11	–
2021	Switch-C [43]	1.57E+12	745GB

Table 1: Overview of recent large language models

...

Bender and Gebru, 2021



Yann LeCun
@ylecun

Current LLMs are trained on text data that would take 20,000 years for a human to read.
And still, they haven't learned that if A is the same as B, then B is the same as A.
Humans get a lot smarter than that with comparatively little training data.

Even corvids, parrots, dogs, and octopuses get smarter than that very, very quickly, with only 2 billion neurons and a few trillion "parameters."

Bias: Timnit Gebru



- Timnit is a leader in AI ethics.
- She joined Google in 2018 but she was fired after she published a research paper.
- The research paper showed that facial recognition is less accurate in identifying people of colour, especially women, which means that it's use is not safe (as can discriminate against them).
- Recommended reads:
- https://en.wikipedia.org/wiki/Timnit_Gebru
- "On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?": https://dl.acm.org/doi/10.1145/3442188.3445922

ChatGPT ethical concerns: Privacy and copyright

- Conversations are stored and used for future training.
- Do not put any personal information.

ChatGPT ethical concerns: Plagiarism and cheating

- Using AI-generated content as original work.
- Generating fake data for research.

- University's blog on Generative AI:

https://blog.westminster.ac.uk/ai/

- Guidance on the use of Generative AI

https://www.westminster.ac.uk/sites/default/public-files/general-documents/GenAI-guidance-for-students.pdf

- University's policy on use of Generative AI:

https://www.westminster.ac.uk/current-students/news/universitys-policy-on-use-of-generative-ai

ChatGPT: Effective use

- As a personalised tutor:

I am a computer science student struggling to understand neural networks. Can you explain neural networks to me?

Expand on the concept of hidden layer.

What is the activation function and why is important?

ChatGPT: Effective use

- Generate ideas

I have implemented a neural network to classify images. How do I know if the network is working?

ChatGPT

Evaluating the performance of a neural network for image classification involves several key steps. Here are some common techniques and metrics to help you assess whether your network is working effectively:

1. **Accuracy:**
   - Calculate the accuracy of your model on a separate test dataset. Accuracy is the ratio of correctly predicted samples to the total number of samples.
   - $\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100\%$
2. **Confusion Matrix:**
   - A confusion matrix provides a more detailed breakdown of the model's performance, showing the number of true positives, true negatives, false positives, and false negatives.
   - It is particularly useful for understanding how well your model is performing across different classes.
3. **Precision, Recall, and F1 Score:**
   - Precision measures the accuracy of positive predictions, while recall measures the fraction of actual positives that were correctly predicted. F1 score is the harmonic mean of precision and recall.
   - $\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$
   - $\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$
   - $\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
4. **Receiver Operating Characteristic (ROC) Curve:**
   - ROC curves are useful for binary classification problems. They illustrate the trade-off between sensitivity (true positive rate) and specificity (true negative rate) at various thresholds.
   - The area under the ROC curve (AUC-ROC) is a commonly used metric.
5. **Mean Squared Error (MSE) or Cross-Entropy Loss:**
   - If your network is designed for regression tasks (e.g., predicting continuous values), you can use mean squared error or cross-entropy loss to evaluate the difference between predicted and true values.
6. **Visual Inspection:**

ChatGPT: Effective use

- Test your knowledge

  Generate 10 multiple-choice questions to test my knowledge on neural networks.

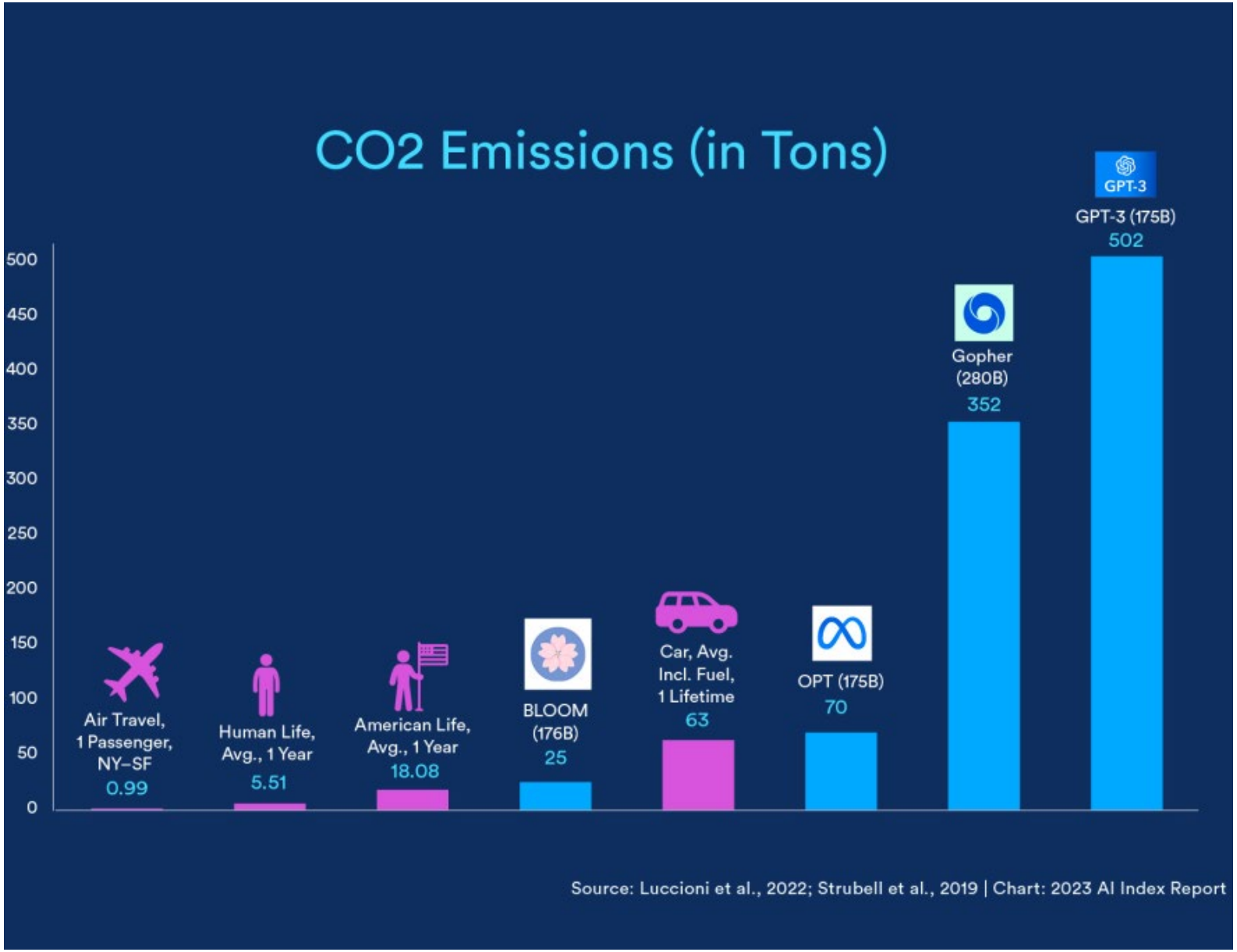
ChatGPT

Certainly! Here are 10 multiple-choice questions to test your knowledge on neural networks:

1. **What is the primary function of the activation function in a neural network?**
   a. Control the learning rate
   b. Introduce non-linearity
   c. Initialize weights
   d. Define the number of hidden layers
2. **Which type of neural network is well-suited for image processing tasks?**
   a. Feedforward Neural Network (FNN)
   b. Recurrent Neural Network (RNN)
   c. Convolutional Neural Network (CNN)
   d. Long Short-Term Memory (LSTM)
3. **What does the vanishing gradient problem refer to in neural networks?**
   a. The model converges too quickly
   b. The model gets stuck in a local minimum
   c. Gradients become too small during backpropagation
   d. Gradients become too large during backpropagation
4. **In L1 regularization, what does the regularization term encourage?**
   a. Large weights
   b. Small weights
   c. Sparsity of weights
   d. Even distribution of weights
5. **What is the purpose of a hidden layer in a neural network?**
   a. Process input data
   b. Make predictions
   c. Adjust weights
   d. Capture and learn complex patterns

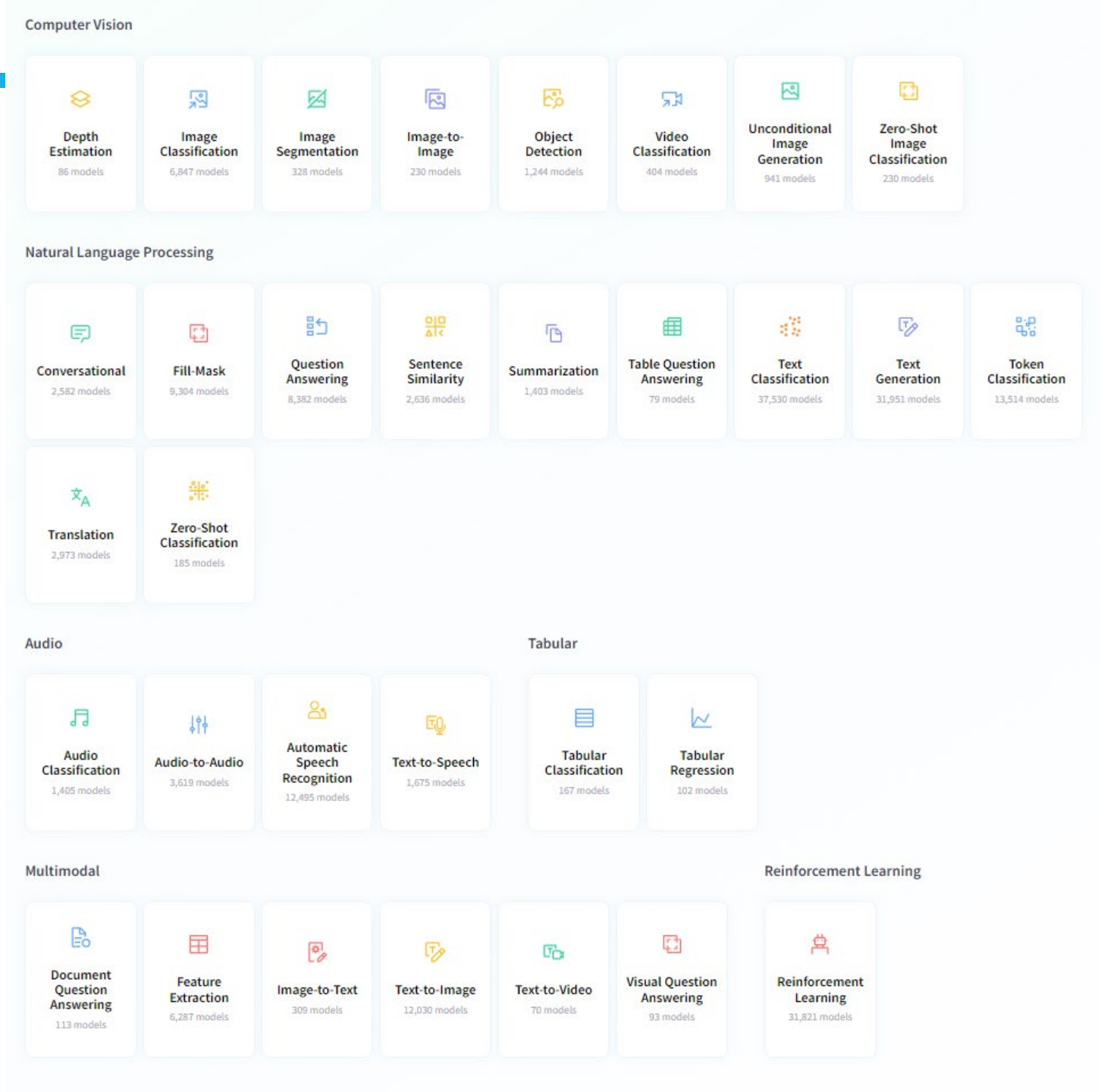
Sustainability

- Environmental impact: LLMs require a large amount of energy to train and run.

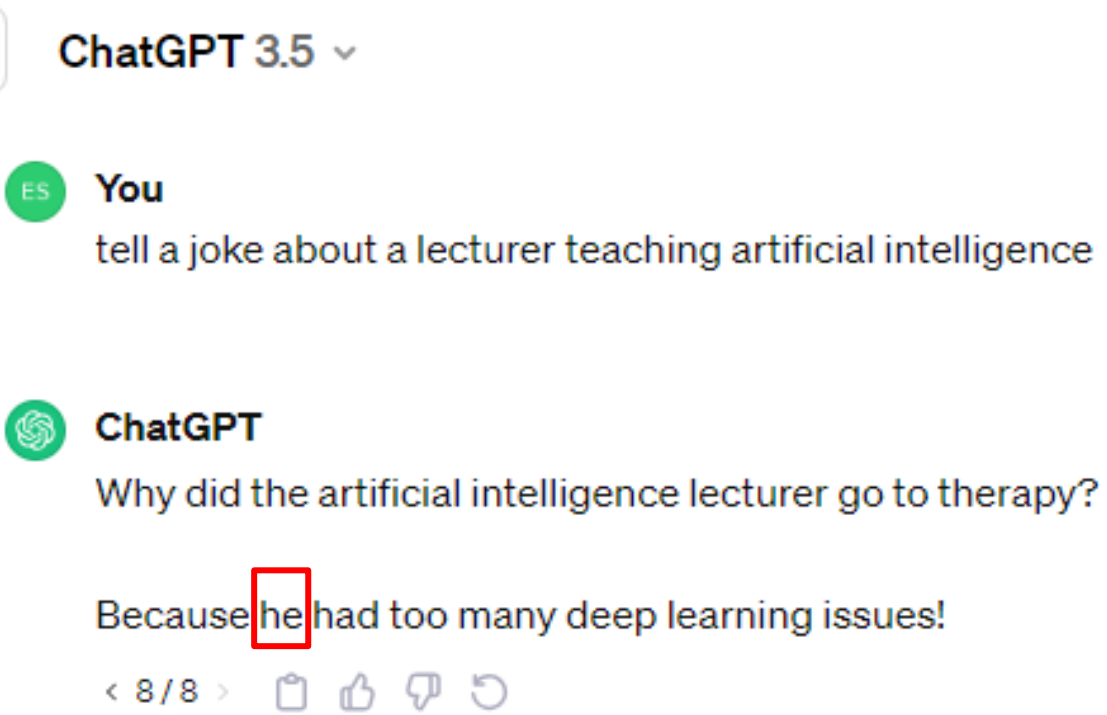


Additional resources

- https://huggingface.co/tasks



Tutorial



THANK YOU!

The AI we have today is the worse we will ever have (it will only get better!).